Research Methodology

Module 5

Data Analysis, Interpretation and Presentation

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1 Data Coding and Data Processing

1.1 Data Processing

Data processing turns **raw, collected data** into a clean, organised form ready for analysis.

Stage	What happens
1. Editing	Check each questionnaire or record for errors, gaps, and inconsistencies.
2. Coding	Assign numbers or symbols to responses.
3. Classification	Group data by common features or into classes.
4. Tabulation	Arrange the data in tables.
5. Data entry	Enter the coded data into software (e.g. a spreadsheet).
6. Cleaning	Remove duplicates, fix typing errors, handle missing values and outliers.

1.2 Editing

- **Field editing:** done soon after collection, while details are fresh.
- **Central (office) editing:** done by the researcher after all forms are collected.
- Check for **completeness, accuracy, consistency, and legibility**.

1.3 Coding

Coding means **converting responses into numerical or symbolic codes**, so a computer can process them. A **codebook** records what each code means.

Variable	Coding example
Gender	1 = Male, 2 = Female, 3 = Other
Satisfaction (Likert)	1 = Very dissatisfied … 5 = Very satisfied
Course year	1 = First, 2 = Second, 3 = Third, 4 = Fourth
Missing answer	A fixed code such as 99, applied consistently

- **Pre-coding:** codes are printed on the questionnaire (closed questions).
- **Post-coding:** codes are made after data collection (open-ended answers are grouped into categories).
- Codes must be **exhaustive** and **mutually exclusive**.
- **Reverse-coded items:** negatively worded questions are flipped (on a 5-point scale, $6 -$ score) before totalling.

1.4 Handling Problems in Data

- **Missing values:** delete the case, replace with the mean or median, or use other estimation methods; decide based on how much is missing.

- **Outliers:** extreme values. Check whether they are errors; do not remove real values just because they are inconvenient.
- **Duplicates and inconsistent entries:** find and correct (e.g. age 5 for a college student).

> **Exam Tip** Always keep a **backup of the raw data** and document every change in the cleaning stage. This keeps the study transparent and replicable.

2 Use of Charts, Tables, and Graphs for Presentation

2.1 Why Present Data Visually?

Good presentation **makes patterns clear**, saves the reader's time, allows quick comparison, and makes findings memorable.

2.2 Tables

Best when **exact values** matter or many variables must be shown together. Give every table a number, a clear title, labelled rows and columns, units, and a source.

2.3 Common Charts and Graphs

Chart	Best used for	Example
Bar chart	Comparing categories	Students per course
Column chart	Same, with vertical bars	Monthly sales
Pie chart	Parts of a whole (few categories)	Share of survey responses
Line graph	Trends over time	Marks across semesters
Histogram	Distribution of continuous data (bars touch)	Spread of exam scores
Scatter plot	Relationship between two variables	Study hours vs. marks
Box plot	Median, spread, outliers	Compare score spread of classes
Pictogram	Simple popular illustration	Counts shown with icons

> **Note** **Bar chart vs. histogram:** a bar chart is for *categories* (gaps between bars); a histogram is for *continuous numeric ranges* (bars touch).

2.4 Rules for Good Presentation

- Choose the chart that suits the **type of data and message**.
- Give a **clear title**, labelled axes with **units**, and a legend if needed.
- Keep it **simple**; avoid 3D effects, clutter, and too many colours.
- Start bar chart axes at **zero**; a truncated axis exaggerates differences.
- Use pie charts only for **few categories** that add up to 100%.

- Mention the **source** and sample size.

> **Watch Out** Misleading graphs (cut axes, uneven scales, cherry-picked ranges) are a form of research dishonesty. Present data to inform, not to persuade unfairly.

3 Interpretation of Statistical Results

3.1 What Is Interpretation?

Interpretation means **explaining what the results mean** in relation to the research questions, hypotheses, and existing literature. Analysis produces numbers; interpretation gives them meaning.

3.2 Steps in Interpretation

1. Restate the research question or hypothesis.
2. Report the key result (test statistic, p-value, effect size).
3. Decide whether the hypothesis is supported or not.
4. Explain the result in plain language and in the study's context.
5. Compare with earlier studies (from the literature review).
6. Mention limitations and possible explanations.

3.3 Interpreting Common Results

Result	How to read it
$p \leq 0.05$	Statistically significant. Evidence against H_0; reject H_0.
$p > 0.05$	Not significant. Not enough evidence to reject H_0 (not proof that H_0 is true).
Confidence interval	Range of plausible values. If it excludes the null value (e.g. 0), the result is significant.
Correlation r	Direction and strength of linear relationship, not causation.
r^2	Proportion of variation in y explained by x.
Regression slope b	Change in y for each one-unit rise in x.
Mean and SD	Typical value and how spread out the data is.

> **Watch Out** **Common errors in interpretation:**
>
> - Treating **correlation as causation**.
> - Confusing **statistical significance** with **practical importance**. A tiny effect can be significant in a very large sample.
> - Generalising beyond the sample or population studied.
> - Ignoring results that do not support the hypothesis.
> - Over-reading a non-significant result as "no effect."

Example. "A one-sample t-test showed that mean daily study time (52 min, $SD = 6$) was not significantly different from 50 min, $t(35) = 2.0$, $p > 0.05$. Therefore, the claim that the mean is 50 minutes is not rejected."

> **Note** Besides p-values, report an **effect size** (such as r, r^2, or the difference in means) so readers know *how large* the effect is, not only *whether* it exists.

4 Drawing Conclusions and Recommendations

4.1 Conclusions

Conclusions are **short, direct answers** to the research questions and objectives, based only on the evidence found.

- Link each conclusion to an **objective or hypothesis**.
- State whether each hypothesis was **supported or not**.
- Stay within the **data**; do not introduce new results or claims.
- Use cautious wording ("suggests," "indicates") where evidence is limited.
- Be concise and clear.

4.2 Recommendations

Recommendations are **practical suggestions** based on the conclusions: what should be done, by whom, and why.

- Must follow **logically from the findings**.
- Should be **specific, realistic, and actionable**.
- Addressed to relevant groups (college management, policy makers, practitioners).

4.3 Limitations and Future Research

- **Limitations:** honest statement of weaknesses such as small sample, one location, self-reported data, or limited time.
- **Scope for future research:** gaps left open, such as larger samples, other populations, or longer duration.

	Conclusion	Recommendation
Nature	What the data shows	What should be done
Basis	Findings	Conclusions
Example	Study time is positively related to marks.	Offer guided study sessions for weaker students.

> **Exam Tip** Structure of a final chapter: **summary of findings → conclusions → recommendations → limitations → future research**.

5 Use of Software Tools for Analysis (Overview)

Software makes analysis **faster, more accurate, and repeatable**, especially for large data sets and complex tests.

Tool	Main use	Remarks
MS Excel	Data entry, tables, charts, basic statistics, simple t-test and regression	Easy, widely available
SPSS	Social science statistics (t-test, ANOVA, chi-square, regression)	Menu-driven, popular in academia
R	Advanced statistics and graphics	Free, open source, needs coding
Python (pandas, SciPy, matplotlib)	Data handling, statistics, visualisation, machine learning	Free, flexible, needs coding
JASP / jamovi	Statistical tests with a simple interface	Free, beginner friendly
Stata, SAS	Professional statistical analysis	Paid, common in economics and industry
Tableau / Power BI	Interactive dashboards and visual reports	Strong visualisation
NVivo / ATLAS.ti	Coding and analysing qualitative data (interviews, text)	For qualitative research
Google Forms	Data collection with auto-charts	Quick surveys

5.1 Typical Workflow

1. Enter or import data (Excel, CSV).
2. Clean and code the data.
3. Run descriptive statistics (mean, SD, frequency).
4. Run suitable tests (t-test, chi-square, ANOVA, correlation, regression).
5. Create tables and charts.
6. Read the output and interpret the results.

5.2 Choosing a Tool

- **Small, simple data:** Excel.
- **Standard statistical tests with menus:** SPSS, JASP, or jamovi.
- **Customised analysis and automation:** R or Python.
- **Dashboards and presentation:** Tableau or Power BI.
- **Text and interview data:** NVivo or ATLAS.ti.

Watch Out Software gives **numbers, not understanding**. Choosing the wrong test or feeding in dirty data produces wrong answers ("garbage in, garbage out"). Always know *why* a test is used and check its assumptions.

6 Quick Revision Sheet

- **Processing:** editing → coding → classification → tabulation → entry → cleaning.
- **Coding:** numeric codes recorded in a codebook; pre-coding vs. post-coding.
- **Charts:** bar (categories), line (trend), pie (parts of whole), histogram (distribution), scatter (relationship).
- **Good graphs:** title, labels, units, simple, zero baseline, source.
- **Interpretation:** explain results in context; $p \leq 0.05$ means significant; report effect size.
- **Avoid:** correlation as causation; significance as importance; over-generalising.
- **Conclusions** answer objectives; **recommendations** suggest action; add limitations and future scope.
- **Software:** Excel, SPSS, R, Python, JASP, Tableau, NVivo; tools help but do not replace understanding.

End of Module 5